

作業一墨西哥帽子

資工三乙 410262107 李佳穎

一、 題目

在墨西哥帽子仿星球旋轉例子中在多放幾頂墨西哥帽子在 3D 中運動(如平移, 旋轉, 或變形如變大變小), (用 OpenGL 函式庫或 WebGL 寫作業皆可以), 鼓勵同學增加創意表現

二、 討論

(一) 基本要求

1. 可以調整地球繞太陽的速度(變快或變慢)
2. 星球旋轉範例內, 將當中的星球改為墨西哥帽子
3. 可以調整月球繞地的速度(變快或變慢)
4. 可以調整眼睛觀看的位置與視線方向
5. 要求同學在墨西哥帽子仿星球旋轉例子中在多放幾頂墨西哥帽子在 3D 中運動(如平移, 旋轉, 或變形如變大變小)

(二) 額外功能

1. 畫面暫停/開始功能
2. 六個帽子皆可調整自轉方向以及自轉速度
3. 將容器設為透明, 加入星空背景
4. 設置方向光源和自然光, 使帽子反射光線
5. 增加流星並可以用按鈕控制顯示及隱藏
6. 六個墨西哥帽子皆可調整顏色
7. 新增音效並可用按鈕控制

(三) 場景設定

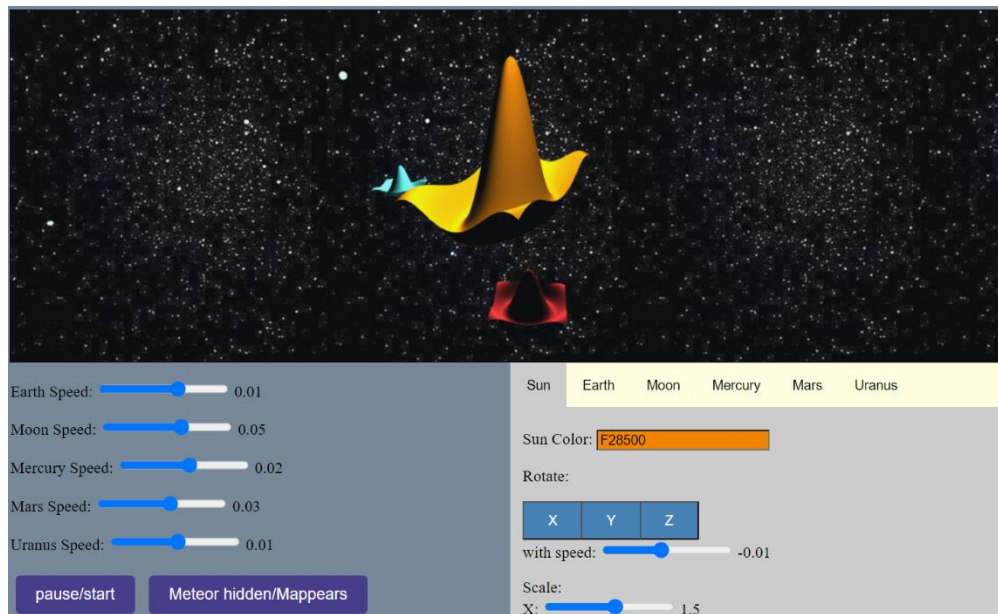
1. 程式碼使用了 Three.js 庫的核心元件, 包括 WebGLRenderer、PerspectiveCamera、Scene 等, 用於建立 3D 渲染器、攝影機和場景
2. 使用 TrackballControls 提供互動性, 使用戶能夠透過滑鼠拖曳旋轉場景
3. 天體的創建與運動: 太陽、地球、月亮、水星、火星和天王星的網格模型使用了 Mesh 和 MeshLambertMaterial。每個天體都有旋轉和位置的更新邏輯, 透過修改其歐拉角和位置向量來模擬運動
4. 流星效果: 使用 createShootingStar 函數創建了流星效果, 每個流星由一個球體和一個長尾巴組成。startShootingStarAnimation 函數實現動畫效果, 不斷更新位置, 並在到達邊界時重置位置, 顯示尾巴效果
5. 動畫循環: 使用 requestAnimationFrame 建立動畫循環, 不斷更新場景中天體的位置與狀態, 實現動態效果
6. 互動和顏色控制: 提供了函數用於切換流星的顯示狀態, 以及暫停

/繼續動畫。使用 jscolor 庫來控制太陽、地球、月亮、水星、火星和天王星的顏色

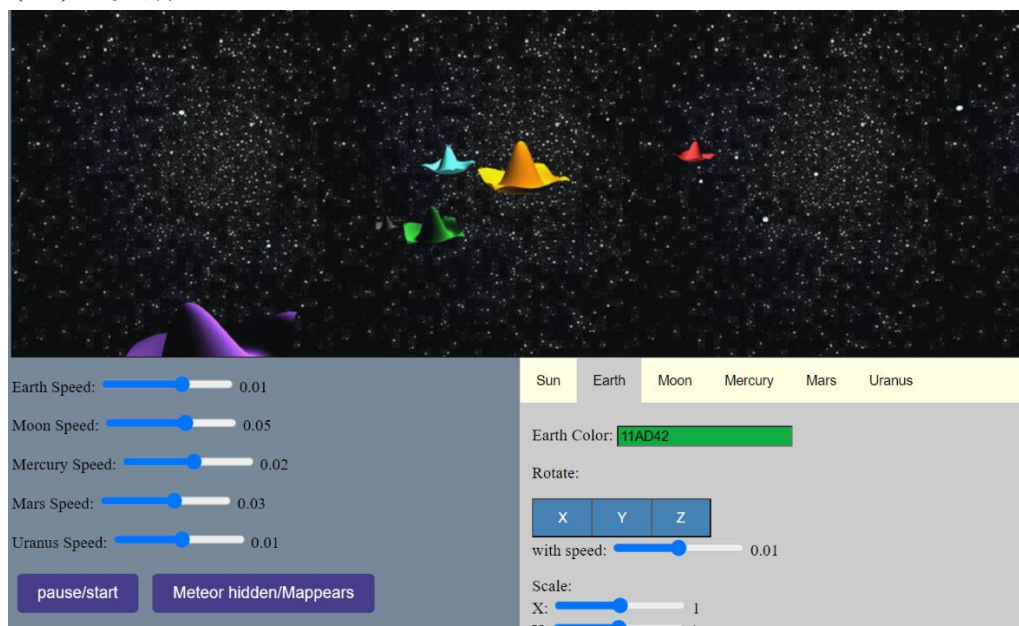
7. 音效：整合音效，包括觸發特定事件時的音樂或效果音。提供音效控制按鈕，讓使用者可以啟用或禁用音效
8. 潛在改進和問題：流星效果的實現可以更靈活，可以考慮使用粒子系統或其他技術來實現更逼真的流星效果

三、 執行畫面

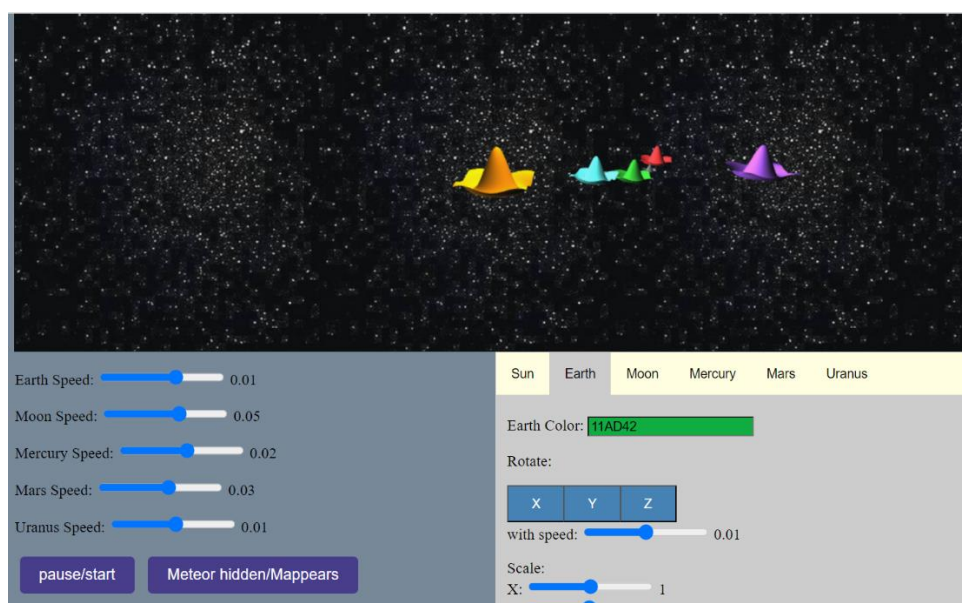
(一) 墨西哥帽子變形



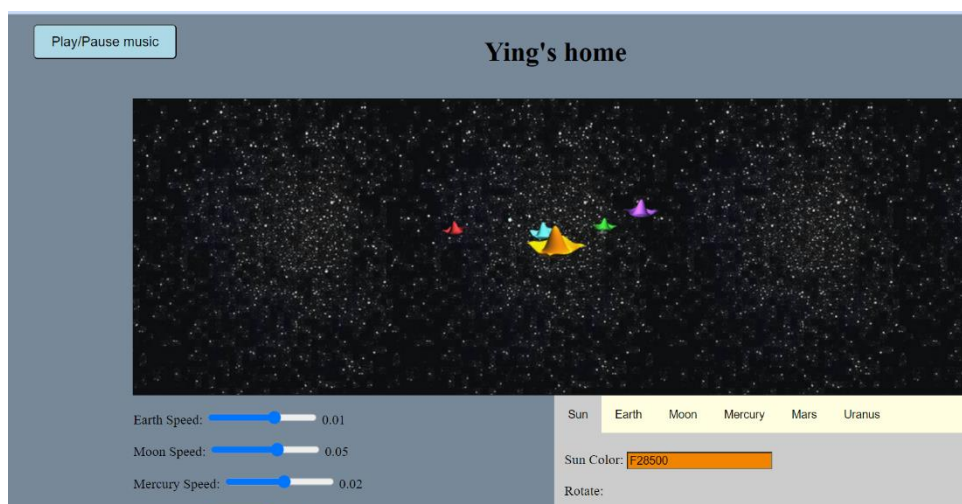
(二) 流星顯示



(三) 流星隱藏



(四) 撥放音樂按鈕



四、 程式碼

(一) Main. js

```
1  'use strict';
2  // set the scene size
3  var WIDTH = 1000, HEIGHT = 350;
4
5  // set some attribute
6  var VIEW_ANGLE = 45;
7  var ASPECT = WIDTH / HEIGHT;
8  var NEAR = 0.1;
9  var FAR = 10000;
10 var isPaused = false;
11 // set the DOM element to attach it
12 var container = document.querySelector('#container');
13
14 // create an webgl renderer, camera and a scene
15 var renderer = new THREE.WebGLRenderer();
16 var camera = new THREE.PerspectiveCamera(
17     VIEW_ANGLE,
18     ASPECT,
19     NEAR,
20     FAR
21 );
22 var scene = new THREE.Scene();
23
24 // add camera(with trackballcontrol) to scene
25 var controls = new THREE.TrackballControls( camera, container );
26 camera.position.z = 500;
27 controls.addEventListener('change', render);
28 scene.add(camera);
29 renderer = new THREE.WebGLRenderer({ antialias: true, alpha: true });
30 renderer.setClearColor(0x000000, 0.0); // 第二个参数是透明度，範圍 0 ( 完全透明 ) 到 1 ( 完全不透明 )
31 renderer.setSize(WIDTH, HEIGHT);
32 var ambientLight = new THREE.AmbientLight( 0x000000 );
33 scene.add( ambientLight );
34
35 var lights = [];
36 lights[ 0 ] = new THREE.PointLight( 0xffffff, 1, 0 );
37 lights[ 1 ] = new THREE.PointLight( 0xffffff, 1, 0 );
38 lights[ 2 ] = new THREE.PointLight( 0xffffff, 1, 0 );
39
40 lights[ 0 ].position.set( 0, 200, 0 );
41 lights[ 1 ].position.set( 100, 200, 100 );
42 lights[ 2 ].position.set( - 100, - 200, - 100 );
43
44 scene.add( lights[ 0 ] );
45 scene.add( lights[ 1 ] );
46 scene.add( lights[ 2 ] );
47
48 // attach the renderer-supplied DOM Element
49 container.appendChild(renderer.domElement);
50
51 // declare global control variable
52 var sun, earth, moon;
53 var speed = [0, 0.01, 0.05];
54 var angle = [0, 0, 0];
55 var rotate = [
56     {speed: 0.01, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false},
57     {speed: 0.01, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false},
58     {speed: 0.01, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false}
59 ];
```

```

60 var Mercury, Mars, Uranus;
61 var newspeed = [0.02, 0.03, 0.01];
62 var newangle = [0, 0, 0];
63 var newrotate = [
64     {speed: 0.03, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false},
65     {speed: 0.02, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false},
66     {speed: 0.01, identity: new THREE.Vector3(0, 1, 0), result: null, lock: false}
67 ];
68 var shootingStars = [];
69 var audio = document.createElement('audio');
70 audio.id = 'backgroundAudio';
71 audio.src = 'js/image/sound.mp3';
72 document.body.appendChild(audio);
73
74 var audioButton = document.createElement('button');
75 audioButton.textContent = 'Play/Pause music';
76 audioButton.addEventListener('click', toggleAudio);
77 audioButton.id = 'customAudioButton'; // 设置按钮的ID
78 document.body.appendChild(audioButton);
79 // 音频播放和暂停的函数
80 function toggleAudio() {
81     if (audio.paused) {
82         audio.play();
83     } else {
84         audio.pause();

```

```

85     }
86 }
87 // 获取按钮元素并设置样式
88 var customAudioButton = document.getElementById('customAudioButton');
89 if (customAudioButton) {
90     customAudioButton.style.position = 'absolute';
91     customAudioButton.style.top = '24px'; // 设置按钮距离顶部的位置
92     customAudioButton.style.left = '20px'; // 设置按钮距离左侧的位置
93     customAudioButton.style.zIndex = '1400'; // 设置按钮的层级
94 }
95
96 function toggleShootingStar(index) {
97     var shootingStar = shootingStars[index];
98     if (shootingStar) {
99         shootingStar.visible = !shootingStar.visible;
100     }
101 }
102 function createShootingStar(x, y, z) {
103     var starGeometry = new THREE.SphereGeometry(2.5);
104     var starMaterial = new THREE.MeshBasicMaterial({ color: 0xe0ffff });
105     var shootingStar = new THREE.Mesh(starGeometry, starMaterial);
106     // 设置流星的初始位置
107     shootingStar.position.set(x, y, z);
108     // 创建尾巴
109
110     var tail = createTail();
111     shootingStar.add(tail);
112     scene.add(shootingStar);
113     shootingStar.tail = tail;
114     // 设置初始尾巴可见性
115     tail.visible = false;
116
117     // 设置流星动画
118     startShootingStarAnimation(shootingStar);
119     shootingStars.push(shootingStar);
120 }
121 function createTail() {
122     var tailGeometry = new THREE.BufferGeometry();
123     var tailMaterial = new THREE.PointsMaterial({
124         color: 0xffff00,
125         size: 10000,
126         transparent: true,

```

```

127         opacity: 1
128     });
129     var tailVertices = [];
130     for (var i = 0; i < 1000; i++) {
131         tailVertices.push(0, 0, 0);
132     }
133     tailGeometry.addAttribute('position', new THREE.Float32BufferAttribute(tailVertices));
134     var tail = new THREE.Line(tailGeometry, tailMaterial);
135
136     // 设置初始尾巴可见性
137     tail.visible = false;
138     return tail;
139 }
140 function startShootingStarAnimation() {
141     // 设置流星动画
142     var animationInterval = setInterval(function () {
143         shootingStar.position.x -= 2; // 修改为适当的速度
144         // 添加其他更新逻辑
145         if (shootingStar.position.x < -WIDTH / 2) {
146             shootingStar.position.x = WIDTH / 2;
147             // 显示尾巴
148             shootingStar.tail.visible = true;
149             // 设置尾巴起始位置
150             shootingStar.tail.position.copy(shootingStar.position);
151         }
152         // 更新尾巴的位置
153         for (var i = 0; i < shootingStar.tail.geometry.attributes.position.array.length; i++) {
154             shootingStar.tail.geometry.attributes.position.array[i] = shootingStar.position.x;
155             shootingStar.tail.geometry.attributes.position.array[i + 1] = shootingStar.position.y;
156             shootingStar.tail.geometry.attributes.position.array[i + 2] = shootingStar.position.z;
157         }
158         shootingStar.tail.geometry.attributes.position.needsUpdate = true;
159         // 隐藏尾巴
160         setTimeout(function () {
161             shootingStar.tail.visible = false;
162         }, 1000); // 尾巴持续时间
163     }, 16); // 帧率约为60fps
164 }
165 function updateShootingStars() {
166     for (var i = 0; i < shootingStars.length; i++) {
167         var shootingStar = shootingStars[i];
168         if (shootingStar.visible) {
169             shootingStar.position.x -= 2;
170
171             if (shootingStar.position.x < -WIDTH / 2) {
172                 shootingStar.position.x = WIDTH / 2;
173                 shootingStar.tail.visible = true;
174                 shootingStar.tail.position.copy(shootingStar.position);
175             }
176
177             for (var j = 0; j < shootingStar.tail.geometry.attributes.position.array.length; j++) {
178                 shootingStar.tail.geometry.attributes.position.array[j] = shootingStar.position.x;
179                 shootingStar.tail.geometry.attributes.position.array[j + 1] = shootingStar.position.y;
180                 shootingStar.tail.geometry.attributes.position.array[j + 2] = shootingStar.position.z;
181             }
182         }
183     }
184 }

```

```

183         shootingStar.tail.geometry.attributes.position.needsUpdate = true;
184
185         setTimeout(function (star) {
186             return function () {
187                 star.tail.visible = false;
188             };
189         })(shootingStar), 1000); // 尾巴持續時間
190     }
191 }
192 }
193
194 function meshCreate () {
195     var sungeometry = new MexicoHat(50.0);
196     var sunmaterial = new THREE.MeshLambertMaterial({color: 0xF28500, side: THREE.DoubleSide});
197     sun = new THREE.Mesh(sungeometry, sunmaterial);
198
199     sun.position.set(0, 0, 0);
200     rotate[0].result = rotate[0].identity.clone();
201     rotate[0].result.multiplyScalar(rotate[0].speed);
202     scene.add(sun);
203
204     /*var loader = new THREE.TextureLoader();
205     loader.crossOrigin = true;
206     var texture = loader.load('http://i.imgur.com/1wZKcwy.jpg');*/
207
208     var earthgeometry = new MexicoHat(25.0);
209     var earthmaterial = new THREE.MeshLambertMaterial({color: 0x228b22, side: THREE.DoubleSide});
210     earth = new THREE.Mesh(earthgeometry, earthmaterial);
211     rotate[1].result = rotate[1].identity.clone();
212     rotate[1].result.multiplyScalar(rotate[1].speed);
213     scene.add(earth);
214
215     var moongeometry = new MexicoHat(10.0);
216     var moonmaterial = new THREE.MeshLambertMaterial({color: 0x444444, side: THREE.DoubleSide});
217     moon = new THREE.Mesh(moongeometry, moonmaterial);
218     rotate[2].result = rotate[2].identity.clone();
219     rotate[2].result.multiplyScalar(rotate[2].speed);
220     scene.add(moon);
221
222
223     var Mercurygeometry = new MexicoHat(30.0);
224     var Mercurymaterial = new THREE.MeshLambertMaterial({color: 0x48d1cc, side: THREE.DoubleSide});
225
226     Mercury = new THREE.Mesh(Mercurygeometry, Mercurymaterial);
227     newrotate[0].result = newrotate[0].identity.clone();
228     newrotate[0].result.multiplyScalar(newrotate[0].speed);
229     scene.add(Mercury);
230
231     /*var loader = new THREE.TextureLoader();
232     loader.crossOrigin = true;
233     var texture = loader.load('http://i.imgur.com/1wZKcwy.jpg');*/
234
235     var Marsgeometry = new MexicoHat(25.0);
236     var Marsmaterial = new THREE.MeshLambertMaterial({color: 0xb22222, side: THREE.DoubleSide});
237     Mars = new THREE.Mesh(Marsgeometry, Marsmaterial);
238     newrotate[1].result = newrotate[1].identity.clone();
239     newrotate[1].result.multiplyScalar(newrotate[1].speed);
240     scene.add(Mars);

```

```

241     var Uranusgeometry = new MexicoHat(40.0);
242     var Uranusmaterial = new THREE.MeshLambertMaterial({color: 0x663399, side: THREE.FrontSide});
243     Uranus = new THREE.Mesh(Uranusgeometry, Uranusmaterial);
244     newrotate[2].result = newrotate[2].identity.clone();
245     newrotate[2].result.multiplyScalar(newrotate[2].speed);
246     scene.add(Uranus);
247     createShootingStar(100, 100, 100);
248     createShootingStar(-200, 50, -200);
249     createShootingStar(300, 150, 300);
250     createShootingStar(50, 0, -20);
251     createShootingStar(-100, 0, 150);
252     createShootingStar(-20, 5, -100);
253     createShootingStar(-100, -10, -20);
254     createShootingStar(300, -100, 0);
255     createShootingStar(300, 100, 100);
256 }
257 // 在 JavaScript 中添加一个函数来切换流星的显示状态
258 function toggleAllShootingStars() {
259     for (var i = 0; i < shootingStars.length; i++) {
260         var shootingStar = shootingStars[i];
261         if (shootingStar) {
262             shootingStar.visible = !shootingStar.visible;
263         }
264     }
265 }
266 function togglePause() {
267     isPaused = !isPaused;
268 }
269 // jscolor callback function
270 function setColor (jscolor) {
271     var targetElement = jscolor.targetElement;
272     switch(targetElement.name){
273         case 'sunC':
274             sun.material.color = new THREE.Color('#' + jscolor);
275             break;
276         case 'earthC':
277             earth.material.color = new THREE.Color('#' + jscolor);
278             break;
279         case 'moonC':
280             moon.material.color = new THREE.Color('#' + jscolor);
281             break;
282         case 'MercuryC':
283             Mercury.material.color = new THREE.Color('#' + jscolor);
284             break;
285         case 'MarsC':
286             Mars.material.color = new THREE.Color('#' + jscolor);
287             break;
288         case 'UranusC':
289             Uranus.material.color = new THREE.Color('#' + jscolor);
290             break;
291     }
292 }

```

```

293 function openTab(evt, tabName) {
294     // Declare all variables
295     var i, tabcontent, tablinks;
296     // Get all elements with class='tabcontent' and hide them
297     tabcontent = document.getElementsByClassName('tabcontent');
298     for (i = 0; i < tabcontent.length; i++) {
299         tabcontent[i].style.display = 'none';
300     }
301     // Get all elements with class='tablinks' and remove the class 'active'
302     tablinks = document.getElementsByClassName('tablinks');
303     for (i = 0; i < tablinks.length; i++) {
304         tablinks[i].className = tablinks[i].className.replace(' active', '');
305     }
306     // Show the current tab, and add an 'active' class to the button that opened the t
307     document.getElementById(tabName).style.display = 'block';
308     evt.currentTarget.className += ' active';
309 }
310 function render () {
311     renderer.render(scene, camera);
312 }
313 function update () {
314     // update control
315     controls.update();
316     // 更新流星的位置或其他属性
317     updateShootingStars();
318     if (!isPaused) {
319         // update position
320         angle[1] += speed[1];
321         angle[2] += speed[2];
322
323         newangle[0] += newspeed[0];
324         newangle[1] += newspeed[1];
325         newangle[2] += newspeed[2];
326
327         sun.rotation.setFromVector3(
328             sun.rotation.toVector3().add(
329                 rotate[0].result
330             )
331         );
332         earth.rotation.setFromVector3(
333             earth.rotation.toVector3().add(
334                 rotate[1].result

```

```

335         )
336     );
337     earth.position.set(
338         200 * Math.cos(angle[1]),
339         0,
340         200 * Math.sin(angle[1])
341     );
342     moon.rotation.setFromVector3(
343         moon.rotation.toVector3().add(
344             rotate[2].result
345         )
346     );
347     moon.position.set(
348         200 * Math.cos(angle[1]) + 50 * Math.cos(angle[2]),
349         0,
350         200 * Math.sin(angle[1]) + 50 * Math.sin(angle[2])

```

```

351     );
352     Mercury.rotation.setFromVector3(
353         Mercury.rotation.toVector3().add(
354             newrotate[0].result
355         )
356     );
357     Mercury.position.set(
358         150 * Math.cos(newangle[0]), 0,
359         100 * Math.sin(newangle[0]),
360     );
361
362     Mars.rotation.setFromVector3(
363         Mars.rotation.toVector3().add(
364             newrotate[1].result
365         )
366     );
367     Mars.position.set(
368         300 * Math.cos(newangle[1]),
369         0,
370         300 * Math.sin(newangle[1])
371     );
372     Uranus.rotation.setFromVector3(
373         Uranus.rotation.toVector3().add(
374             newrotate[2].result
375         )
376     );
377     Uranus.position.set(
378         400 * Math.cos(newangle[2]),
379         0,
380         400 * Math.sin(newangle[2])
381     );
382     sun.material.needsUpdate = true;
383     sun.geometry.uvsNeedUpdate = true;
384     sun.geometry.buffersNeedUpdate = true;
385
386     // Draw!
387     render();
388 }
389 // Schedule the next frame.
390 requestAnimationFrame(update);
391 }
392 // Schedule the first frame.

```

```

393 meshCreate ();
394 listener ();
395 update ();
396 document.querySelector('.tablins').click();

```

(二) listeners. js

```

1  function listener () {
2      // declare DOM list
3      var earthSpeedDOM = document.querySelector('.stats span.earthS');
4      var moonSpeedDOM  = document.querySelector('.stats span.moonS');
5      var earthInputDOM = document.querySelector('.stats input.earthS');
6      var moonInputDOM  = document.querySelector('.stats input.moonS');
7      var sunColorSel   = document.querySelector('.stats input[name="sunC"]');
8      var earthColorSel = document.querySelector('.stats input[name="earthC"]');
9      var moonColorSel  = document.querySelector('.stats input[name="moonC"]');
10     var sunRotateSel  = document.querySelector('.sunR').getElementsByTagName("button");
11     var sunRotateDOM  = document.querySelectorAll('.sunRS');

```

```

12  var earthRotateSel = document.querySelector('.earthR').getElementsByName("butto
13  var earthRotateDOM = document.querySelectorAll('.earthRS');
14  var moonRotateSel = document.querySelector('.moonR').getElementsByName("button"
15  var moonRotateDOM = document.querySelectorAll('.moonRS');
16
17  var MercuryRotateSel = document.querySelector('.MercuryR').getElementsByName("b
18  var MercuryRotateDOM = document.querySelectorAll('.MercuryRS');
19  var MercurySpeedDOM = document.querySelector('.stats span.MercuryS');
20  var MercuryInputDOM = document.querySelector('.stats input.MercuryS');
21  var MercuryColorSel = document.querySelector('.stats input[name="MercuryC"]');
22
23  var MarsRotateSel = document.querySelector('.MarsR').getElementsByName("button"
24  var MarsRotateDOM = document.querySelectorAll('.MarsRS');
25  var MarsSpeedDOM = document.querySelector('.stats span.MarsS');
26  var MarsInputDOM = document.querySelector('.stats input.MarsS');
27  var MarsColorSel = document.querySelector('.stats input[name="MarsC"]');
28
29  var UranusRotateSel = document.querySelector('.UranusR').getElementsByName("but
30  var UranusRotateDOM = document.querySelectorAll('.UranusRS');
31  var UranusSpeedDOM = document.querySelector('.stats span.UranusS');
32  var UranusInputDOM = document.querySelector('.stats input.UranusS');
33  var UranusColorSel = document.querySelector('.stats input[name="UranusC"]');
34
35  var sunScaleSel = document.querySelectorAll('.sunSC');
36  var earthScaleSel = document.querySelectorAll('.earthSC');
37  var moonScaleSel = document.querySelectorAll('.moonSC');
38  var MercuryScaleSel = document.querySelectorAll('.MercurySC');
39  var MarsScaleSel = document.querySelectorAll('.MarsSC');
40  var UranusScaleSel = document.querySelectorAll('.UranusSC');
41  // setup initial value
42
43  earthSpeedDOM.innerText = speed[1];
44  moonSpeedDOM.innerText = speed[2];
45  MercurySpeedDOM.innerText = newspeed[0];
46  MarsSpeedDOM.innerText = newspeed[1];
47  UranusSpeedDOM.innerText = newspeed[2];

```

```

48
49  earthInputDOM.value = speed[1];
50  moonInputDOM.value = speed[2];
51  MercuryInputDOM.value = newspeed[0];
52  MarsInputDOM.value = newspeed[1];
53  UranusInputDOM.value = newspeed[2];
54  sunColorSel.value = 'F28500';
55  earthColorSel.value = '11AD42';
56  moonColorSel.value = '444444';
57  var toggleStarButton = document.getElementById('toggleStarButton');
58  toggleStarButton.addEventListener('click', function () {
59      toggleAllShootingStars();
60  });
61

```

```

62  var pauseButton = document.getElementById('pauseButton');
63
64  pauseButton.addEventListener('click', function () {
65      togglePause(); // 调用暂停/播放函数
66  });
67

```

```

68 // setup input listener
69 earthInputDOM.addEventListener('input', function (e) {
70     speed[1] = parseFloat(e.target.value);
71     earthSpeedDOM.innerText = speed[1];
72 });
73 moonInputDOM.addEventListener('input', function (e) {
74     speed[2] = parseFloat(e.target.value);
75     moonSpeedDOM.innerText = speed[2];
76 });
77 MercuryInputDOM.addEventListener('input', function (e) {
78     newspeed[0] = parseFloat(e.target.value);
79     MercurySpeedDOM.innerText = newspeed[0];
80 });
81 MarsInputDOM.addEventListener('input', function (e) {
82     newspeed[1] = parseFloat(e.target.value);
83     MarsSpeedDOM.innerText = newspeed[1];
84 });
85 UranusInputDOM.addEventListener('input', function (e) {
86     newspeed[2] = parseFloat(e.target.value);
87     UranusSpeedDOM.innerText = newspeed[2];
88 });
89
90 for(var i = 0; i < sunRotateSel.length; i++){
91     sunRotateSel[i].addEventListener('click', function(e){
92         switch(e.target.innerText){
93             case 'X': rotate[0].identity = new THREE.Vector3(1, 0, 0);break;
94             case 'Y': rotate[0].identity = new THREE.Vector3(0, 1, 0);break;
95             case 'Z': rotate[0].identity = new THREE.Vector3(0, 0, 1);break;
96         }
97         rotate[0].result = rotate[0].identity;
98         rotate[0].result.multiplyScalar(rotate[0].speed);
99     });
100 }

```

```

101 sunRotateDOM[1].innerText = rotate[0].speed;
102 sunRotateDOM[0].addEventListener('input', function(e) {
103     if(rotate[0].lock) return;
104     rotate[0].lock = true;
105     rotate[0].speed = parseFloat(e.target.value);
106     sunRotateDOM[1].innerText = rotate[0].speed;
107     rotate[0].result = rotate[0].identity.clone();
108     rotate[0].result.multiplyScalar(rotate[0].speed);
109     rotate[0].lock = false;
110 });
111 for(var i = 0; i < earthRotateSel.length; i++){
112     earthRotateSel[i].addEventListener('click', function(e){
113         switch(e.target.innerText){
114             case 'X': rotate[1].identity = new THREE.Vector3(1, 0, 0);break;
115             case 'Y': rotate[1].identity = new THREE.Vector3(0, 1, 0);break;
116             case 'Z': rotate[1].identity = new THREE.Vector3(0, 0, 1);break;
117         }
118         rotate[1].result = rotate[1].identity;
119         rotate[1].result.multiplyScalar(rotate[1].speed);
120     });
121 }

```

```

122     earthRotateDOM[1].innerText = rotate[1].speed;
123     earthRotateDOM[0].addEventListener('input', function(e) {
124         if(rotate[1].lock) return;
125         rotate[1].lock = true;
126         rotate[1].speed = parseFloat(e.target.value);
127         earthRotateDOM[1].innerText = rotate[1].speed;
128         rotate[1].result = rotate[1].identity.clone();
129         rotate[1].result.multiplyScalar(rotate[1].speed);
130         rotate[1].lock = false;
131     });
132     for(var i = 0; i < moonRotateSel.length; i++){
133         moonRotateSel[i].addEventListener('click', function(e){
134             switch(e.target.innerText){
135                 case 'X': rotate[2].identity = new THREE.Vector3(1, 0, 0);break;
136                 case 'Y': rotate[2].identity = new THREE.Vector3(0, 1, 0);break;
137                 case 'Z': rotate[2].identity = new THREE.Vector3(0, 0, 1);break;
138             }
139             rotate[2].result = rotate[2].identity;
140             rotate[2].result.multiplyScalar(rotate[2].speed);
141         });
142     }
143     moonRotateDOM[1].innerText = rotate[2].speed;
144     moonRotateDOM[0].addEventListener('input', function(e) {
145         if(rotate[2].lock) return;
146         rotate[2].lock = true;
147         rotate[2].speed = parseFloat(e.target.value);
148         moonRotateDOM[1].innerText = rotate[2].speed;
149         rotate[2].result = rotate[2].identity.clone();
150         rotate[2].result.multiplyScalar(rotate[2].speed);
151         rotate[2].lock = false;
152     });
153
154     for (var i = 0; i < MercuryRotateSel.length; i++) {
155         MercuryRotateSel[i].addEventListener('click', function (e) {
156             switch (e.target.innerText) {
157                 case 'X': newrotate[0].identity = new THREE.Vector3(1, 0, 0); break;
158                 case 'Y': newrotate[0].identity = new THREE.Vector3(0, 1, 0); break;
159                 case 'Z': newrotate[0].identity = new THREE.Vector3(0, 0, 1); break;
160             }
161             newrotate[0].result = newrotate[0].identity;
162             newrotate[0].result.multiplyScalar(newrotate[0].speed);
163         });

```

```

164     }
165     MercuryRotateDOM[1].innerText = newrotate[0].speed;
166     MercuryRotateDOM[0].addEventListener('input', function (e) {
167         if (newrotate[0].lock) return;
168         newrotate[0].lock = true;
169         newrotate[0].speed = parseFloat(e.target.value);
170         MercuryRotateDOM[1].innerText = newrotate[0].speed;
171         newrotate[0].result = newrotate[0].identity.clone();
172         newrotate[0].result.multiplyScalar(newrotate[0].speed);
173         newrotate[0].lock = false;
174     });
175

```

```

176     for (var i = 0; i < MarsRotateSel.length; i++) {
177         MarsRotateSel[i].addEventListener('click', function (e) {
178             switch (e.target.innerText) {
179                 case 'X': newrotate[1].identity = new THREE.Vector3(1, 0, 0); break;
180                 case 'Y': newrotate[1].identity = new THREE.Vector3(0, 1, 0); break;
181                 case 'Z': newrotate[1].identity = new THREE.Vector3(0, 0, 1); break;
182             }
183             newrotate[1].result = newrotate[1].identity;
184             newrotate[1].result.multiplyScalar(newrotate[1].speed);
185         });
186     }
187     MarsRotateDOM[1].innerText = newrotate[1].speed;
188     MarsRotateDOM[0].addEventListener('input', function (e) {
189         if (newrotate[1].lock) return;
190         newrotate[1].lock = true;
191         newrotate[1].speed = parseFloat(e.target.value);
192         MarsRotateDOM[1].innerText = newrotate[1].speed;
193         newrotate[1].result = newrotate[1].identity.clone();
194         newrotate[1].result.multiplyScalar(newrotate[1].speed);
195         newrotate[1].lock = false;
196     });
197
198     for (var i = 0; i < UranusRotateSel.length; i++) {
199         UranusRotateSel[i].addEventListener('click', function (e) {
200             switch (e.target.innerText) {
201                 case 'X': newrotate[2].identity = new THREE.Vector3(1, 0, 0); break;
202                 case 'Y': newrotate[2].identity = new THREE.Vector3(0, 1, 0); break;
203                 case 'Z': newrotate[2].identity = new THREE.Vector3(0, 0, 1); break;
204             }
205             newrotate[2].result = newrotate[2].identity;
206             newrotate[2].result.multiplyScalar(newrotate[2].speed);
207         });
208     }
209     UranusRotateDOM[1].innerText = newrotate[1].speed;
210     UranusRotateDOM[0].addEventListener('input', function (e) {
211         if (newrotate[2].lock) return;
212         newrotate[2].lock = true;
213         newrotate[2].speed = parseFloat(e.target.value);
214         UranusRotateDOM[1].innerText = newrotate[2].speed;
215         newrotate[2].result = newrotate[2].identity.clone();
216         newrotate[2].result.multiplyScalar(newrotate[2].speed);
217         newrotate[2].lock = false;

```

```

218     });
219
220     sunScaleSel[1].innerText = sun.scale.x;
221     sunScaleSel[0].value = Math.log(sun.scale.x) / Math.log(1.5);
222     sunScaleSel[0].addEventListener('input', function(e){
223         sun.scale.x = Math.pow(1.5, parseFloat(e.target.value));
224         e.target.nextSibling.nextSibling.innerText = sun.scale.x;
225     });
226     sunScaleSel[3].innerText = sun.scale.y;
227     sunScaleSel[2].value = Math.log(sun.scale.y) / Math.log(1.5);
228     sunScaleSel[2].addEventListener('input', function(e){
229         sun.scale.y = Math.pow(1.5, parseFloat(e.target.value));
230         e.target.nextSibling.nextSibling.innerText = sun.scale.y;
231     });
232     sunScaleSel[5].innerText = sun.scale.z;
233     sunScaleSel[4].value = Math.log(sun.scale.z) / Math.log(1.5);
234     sunScaleSel[4].addEventListener('input', function(e){
235         sun.scale.z = Math.pow(1.5, parseFloat(e.target.value));

```

```

236     e.target.nextSibling.nextSibling.innerText = sun.scale.z;
237 });
238
239 earthScaleSel[1].innerText = earth.scale.x;
240 earthScaleSel[0].value = Math.log(earth.scale.x) / Math.log(1.5);
241 earthScaleSel[0].addEventListener('input', function(e){
242     earth.scale.x = Math.pow(1.5, parseFloat(e.target.value));
243     e.target.nextSibling.nextSibling.innerText = earth.scale.x;
244 });
245 earthScaleSel[3].innerText = earth.scale.y;
246 earthScaleSel[2].value = Math.log(earth.scale.y) / Math.log(1.5);
247 earthScaleSel[2].addEventListener('input', function(e){
248     earth.scale.y = Math.pow(1.5, parseFloat(e.target.value));
249     e.target.nextSibling.nextSibling.innerText = earth.scale.y;
250 });
251 earthScaleSel[5].innerText = earth.scale.z;
252 earthScaleSel[4].value = Math.log(earth.scale.z) / Math.log(1.5);
253 earthScaleSel[4].addEventListener('input', function(e){
254     earth.scale.z = Math.pow(1.5, parseFloat(e.target.value));
255     e.target.nextSibling.nextSibling.innerText = earth.scale.z;
256 });
257
258 moonScaleSel[1].innerText = moon.scale.x;
259 moonScaleSel[0].value = Math.log(moon.scale.x) / Math.log(1.5);
260 moonScaleSel[0].addEventListener('input', function(e){
261     moon.scale.x = Math.pow(1.5, parseFloat(e.target.value));
262     e.target.nextSibling.nextSibling.innerText = moon.scale.x;
263 });
264 moonScaleSel[3].innerText = moon.scale.y;
265 moonScaleSel[2].value = Math.log(moon.scale.y) / Math.log(1.5);
266 moonScaleSel[2].addEventListener('input', function(e){
267     moon.scale.y = Math.pow(1.5, parseFloat(e.target.value));
268     e.target.nextSibling.nextSibling.innerText = moon.scale.y;
269 });
270 moonScaleSel[5].innerText = moon.scale.z;
271 moonScaleSel[4].value = Math.log(moon.scale.z) / Math.log(1.5);
272 moonScaleSel[4].addEventListener('input', function(e){
273     moon.scale.z = Math.pow(1.5, parseFloat(e.target.value));
274     e.target.nextSibling.nextSibling.innerText = moon.scale.z;
275 });
276
277 MercuryScaleSel[1].innerText = Mercury.scale.x;
278 MercuryScaleSel[0].value = Math.log(Mercury.scale.x) / Math.log(1.5);
279 MercuryScaleSel[0].addEventListener('input', function (e) {
280     Mercury.scale.x = Math.pow(1.5, parseFloat(e.target.value));
281     e.target.nextSibling.nextSibling.innerText = Mercury.scale.x;
282 });
283 MercuryScaleSel[3].innerText = Mercury.scale.y;
284 MercuryScaleSel[2].value = Math.log(Mercury.scale.y) / Math.log(1.5);
285 MercuryScaleSel[2].addEventListener('input', function (e) {
286     Mercury.scale.y = Math.pow(1.5, parseFloat(e.target.value));
287     e.target.nextSibling.nextSibling.innerText = Mercury.scale.y;
288 });
289 MercuryScaleSel[5].innerText = Mercury.scale.z;
290 MercuryScaleSel[4].value = Math.log(Mercury.scale.z) / Math.log(1.5);
291 MercuryScaleSel[4].addEventListener('input', function (e) {
292     Mercury.scale.z = Math.pow(1.5, parseFloat(e.target.value));
293     e.target.nextSibling.nextSibling.innerText = Mercury.scale.z;
294 });
295

```

```

296     MarsScaleSel[1].innerText = Mars.scale.x;
297     MarsScaleSel[0].value = Math.log(Mars.scale.x) / Math.log(1.5);
298     MarsScaleSel[0].addEventListener('input', function (e) {
299         Mars.scale.x = Math.pow(1.5, parseFloat(e.target.value));
300         e.target.nextSibling.nextSibling.innerText = Mars.scale.x;
301     });
302     MarsScaleSel[3].innerText = Mars.scale.y;
303     MarsScaleSel[2].value = Math.log(Mars.scale.y) / Math.log(1.5);
304     MarsScaleSel[2].addEventListener('input', function (e) {
305         Mars.scale.y = Math.pow(1.5, parseFloat(e.target.value));
306         e.target.nextSibling.nextSibling.innerText = Mars.scale.y;
307     });
308     MarsScaleSel[5].innerText = Mars.scale.z;
309     MarsScaleSel[4].value = Math.log(Mars.scale.z) / Math.log(1.5);
310     MarsScaleSel[4].addEventListener('input', function (e) {
311         Mars.scale.z = Math.pow(1.5, parseFloat(e.target.value));
312         e.target.nextSibling.nextSibling.innerText = Mars.scale.z;
313     });
314     UranusScaleSel[1].innerText = Uranus.scale.x;
315     UranusScaleSel[0].value = Math.log(Uranus.scale.x) / Math.log(1.5);
316     UranusScaleSel[0].addEventListener('input', function (e) {
317         Uranus.scale.x = Math.pow(1.5, parseFloat(e.target.value));
318         e.target.nextSibling.nextSibling.innerText = Uranus.scale.x;
319     });
320     UranusScaleSel[3].innerText = Uranus.scale.y;
321     UranusScaleSel[2].value = Math.log(Uranus.scale.y) / Math.log(1.5);
322     UranusScaleSel[2].addEventListener('input', function (e) {
323         Uranus.scale.y = Math.pow(1.5, parseFloat(e.target.value));
324         e.target.nextSibling.nextSibling.innerText = Uranus.scale.y;
325     });
326     UranusScaleSel[5].innerText = Uranus.scale.z;
327     UranusScaleSel[4].value = Math.log(Uranus.scale.z) / Math.log(1.5);
328     UranusScaleSel[4].addEventListener('input', function (e) {
329         Uranus.scale.z = Math.pow(1.5, parseFloat(e.target.value));
330         e.target.nextSibling.nextSibling.innerText = Uranus.scale.z;
331     });
332 }

```

(三) Index.html

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>作業一墨西哥帽子</title>
6      <link rel="stylesheet" href="style.css">
7  </head>
8  <body>
9      <h1 class="title">Ying's home</h1>
10     <
11     <div id="container"></div>
12     <div class="stats">
13         <div class="col2">
14             <p>
15                 Earth Speed:
16                 <input class="earthS" type="range" min="-0.04" max="0.04" step="0.001"
17                 <span class="earthS">0</span>
18             </p>
19             <p>
20                 Moon Speed:
21                 <input class="moonS" type="range" min="-0.2" max="0.2" step="0.005" />

```

```

22         <span class="moonS">0</span>
23     </p>
24     <p>
25         Mercury Speed:
26         <input class="MercuryS" type="range" min="-0.2" max="0.2" step="0.005"
27         <span class="MercuryS">0</span>
28     </p>
29     <p>
30         Mars Speed:
31         <input class="MarsS" type="range" min="-0.2" max="0.2" step="0.005" />
32         <span class="MarsS">0</span>
33     </p>
34     <p>
35         Uranus Speed:
36         <input class="UranusS" type="range" min="-0.2" max="0.2" step="0.005" />
37         <span class="UranusS">0</span>
38     </p>
39     <p>
40         <button id="pauseButton">pause/start</button>
41         <button id="toggleStarButton">Meteor hidden/Mappears</button>
42     </p>
43 </div>
44 <div class="col2">
45     <div class="tab">
46         <button class="tablinks" onclick="openTab(event, 'SunTab')">Sun</button>
47         <button class="tablinks" onclick="openTab(event, 'EarthTab')">Earth</button>
48         <button class="tablinks" onclick="openTab(event, 'MoonTab')">Moon</button>
49         <button class="tablinks" onclick="openTab(event, 'MercuryTab')">Mercury</button>
50         <button class="tablinks" onclick="openTab(event, 'MarsTab')">Mars</button>
51         <button class="tablinks" onclick="openTab(event, 'UranusTab')">Uranus</button>
52     </div>
53
54     <div id="SunTab" class="tabcontent">
55         <p>
56             Sun Color:
57             <input name="sunC"
58             class="jscolor {onFineChange:'setColor(this)'}"
59             value="EAF292">
60         </p>
61         <p>
62             Rotate:
63             <div class="btn-group sunR">
64                 <button>X</button>
65                 <button>Y</button>
66                 <button>Z</button>
67             </div>
68             with speed:
69             <input class="sunRS" type="range" min="-0.1" max="0.1" step="0.01"
70             <span class="sunRS">0</span>
71         </p>
72         <p>
73             Scale: <br />
74             X: <input class="sunSC" type="range" min="-10" max="10" step="1" />
75             <span class="sunSC">1</span><br />
76             Y: <input class="sunSC" type="range" min="-10" max="10" step="1" />
77             <span class="sunSC">1</span><br />
78             Z: <input class="sunSC" type="range" min="-10" max="10" step="1" />
79             <span class="sunSC">1</span><br />
80         </p>
81     </div>

```

```

82         <div id="EarthTab" class="tabcontent">
83             <p>
84                 Earth Color:
85                 <input name="earthC"
86                     class="jscolor {onFineChange:'setColor(this)'}"
87                     value="cc66ff">
88             </p>
89             <p>
90                 Rotate:
91                 <div class="btn-group earthR">
92                     <button>X</button>
93                     <button>Y</button>
94                     <button>Z</button>
95                 </div>
96                 with speed:
97                 <input class="earthRS" type="range" min="-0.1" max="0.1" step="0.01" />
98                 <span class="earthRS">0</span>
99             </p>
100             <p>
101                 Scale: <br />
102                 X: <input class="earthSC" type="range" min="-10" max="10" step="1" />
103                 <span class="earthSC">1</span><br />
104                 Y: <input class="earthSC" type="range" min="-10" max="10" step="1" />
105                 <span class="earthSC">1</span><br />
106                 Z: <input class="earthSC" type="range" min="-10" max="10" step="1" />
107                 <span class="earthSC">1</span><br />
108             </p>
109         </div>
110         <div id="MoonTab" class="tabcontent">
111             <p>
112                 Moon Color:
113                 <input name="moonC"
114                     class="jscolor {onFineChange:'setColor(this)'}"
115                     value="c0c0cf">
116             </p>
117             <p>
118                 Rotate:
119                 <div class="btn-group moonR">
120                     <button>X</button>
121                     <button>Y</button>
122                     <button>Z</button>
123                 </div>
124                 with speed:
125                 <input class="moonRS" type="range" min="-0.1" max="0.1" step="0.01" />
126                 <span class="moonRS">0</span>
127             </p>
128             <p>
129                 Scale: <br />
130                 X: <input class="moonSC" type="range" min="-10" max="10" step="1" />
131                 <span class="moonSC">1</span><br />
132                 Y: <input class="moonSC" type="range" min="-10" max="10" step="1" />
133                 <span class="moonSC">1</span><br />
134                 Z: <input class="moonSC" type="range" min="-10" max="10" step="1" />
135                 <span class="moonSC">1</span><br />
136             </p>
137         </div>
138         <div id="MercuryTab" class="tabcontent">
139             <p>
140                 Mercury Color:
141                 <input name="MercuryC"

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```

142         class="jscolor {onFineChange:'setColor(this)'}"
143         value="48d1cc">
144     </p>
145     <p>
146         Rotate:
147         <div class="btn-group MercuryR">
148             <button>X</button>
149             <button>Y</button>
150             <button>Z</button>
151         </div>
152         with speed:
153         <input class="MercuryRS" type="range" min="-0.1" max="0.1" step="0
154         <span class="MercuryRS">0</span>
155     </p>
156     <p>
157         Scale: <br />
158         X: <input class="MercurySC" type="range" min="-10" max="10" step="
159         <span class="MercurySC">1</span><br />
160         Y: <input class="MercurySC" type="range" min="-10" max="10" step="
161         <span class="MercurySC">1</span><br />
162         Z: <input class="MercurySC" type="range" min="-10" max="10" step="
163         <span class="MercurySC">1</span><br />
164     </p>
165 </div>
166 <div id="MarsTab" class="tabcontent">
167     <p>
168         Mars Color:
169         <input name="MarsC"
170             class="jscolor {onFineChange:'setColor(this)'}"
171             value="b22222">
172     </p>
173     <p>
174         Rotate:
175         <div class="btn-group MarsR">
176             <button>X</button>
177             <button>Y</button>
178             <button>Z</button>
179         </div>
180         with speed:
181         <input class="MarsRS" type="range" min="-0.1" max="0.1" step="0.01
182         <span class="MarsRS">0</span>
183     </p>

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```

184     <p>
185         Scale: <br />
186         X: <input class="MarsSC" type="range" min="-10" max="10" step="1"
187         <span class="MarsSC">1</span><br />
188         Y: <input class="MarsSC" type="range" min="-10" max="10" step="1"
189         <span class="MarsSC">1</span><br />
190         Z: <input class="MarsSC" type="range" min="-10" max="10" step="1"
191         <span class="MarsSC">1</span><br />
192     </p>
193 </div>
194 <div id="UranusTab" class="tabcontent">
195     <p>
196         Uranus Color:
197         <input name="UranusC"
198             class="jscolor {onFineChange:'setColor(this)'}"
199             value="663399">

```

```

200         </p>
201         <p>
202             Rotate:
203             <div class="btn-group UranusR">
204                 <button>X</button>
205                 <button>Y</button>
206                 <button>Z</button>
207             </div>
208             with speed:
209             <input class="UranusRS" type="range" min="-0.1" max="0.1" step="0.
210             <span class="UranusRS">0</span>
211         </p>
212         <p>
213             Scale: <br />
214             X: <input class="UranusSC" type="range" min="-10" max="10" step="1
215             <span class="UranusSC">1</span><br />
216             Y: <input class="UranusSC" type="range" min="-10" max="10" step="1
217             <span class="UranusSC">1</span><br />
218             Z: <input class="UranusSC" type="range" min="-10" max="10" step="1
219             <span class="UranusSC">1</span><br />
220         </p>
221     </div>
222 </div>
223 </div>
224 <script src="js/three.js"></script>
225 <script src="js/TrackballControls.js"></script>
226 <script src="js/vendor/jscolor.min.js"></script>
227 <script src="js/meshList/mexicoHat.js"></script>
228 <script src="js/listeners.js"></script>
229 <script src="js/main.js"></script>
230 <body style="background-color: #778899; ">
231 </html>

```